



The Panama Canal

Navigating Sustainability Challenges

Case Study

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The Panama Canal: Navigating Sustainability Challenges

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*"Climate change is reshaping countries' economic and trade prospects, and is a major threat to future growth and prosperity. Higher temperatures, rising sea levels and more frequent extreme weather events bring the prospect of productivity losses, production shortages, damaged transport infrastructure, and supply disruptions."*¹

– World Trade Report 2022

*"Global trade is very dependent on environmental changes and our social practices. It shows just how fragile our supply chain is. Climate issues in one country can potentially disrupt the entire world's supply chain."*²

– Ayman Omar³

*"Climate change is a slow-moving crisis that is going to last a very, very long time, and it's going to require some fundamental changes. Every coastal community, every coastal transportation network is going to face some risks from this, and we're not going to have nearly enough resources to make all the investments that are required."*⁴

– Austin Becker⁵

In the intricate web of global commerce, supply chains acted as the lifeblood of economic production, intricately linking nations and industries. Valued at nearly \$20 trillion annually, these networks facilitated the seamless exchange of goods across borders, powered by a myriad of components sourced from around the world. Yet, their resilience faced a formidable test when the COVID-19 pandemic and escalating climate-related disasters revealed their vulnerabilities. As the pandemic raged, and extreme weather events intensified due to climate change, the fragility of these supply chains became increasingly apparent. East Asia and the Pacific bore witness to the grim reality, as natural catastrophes loomed large on the horizon. The World Economic Forum sounded the alarm, highlighting the looming threats of extreme

¹ Feingold Spencer, "'Unprecedented challenges': Drought at the Panama Canal disrupts global trade", <https://www.weforum.org/agenda/2023/08/drought-at-the-panama-canal-poses-unprecedented-challenges-to-global-trade/>, August 22nd 2023

² McCrary Jamie, "Panama Canal Droughts Expose Why Climate Change Is a Global Trade Issue, Too", <https://kogod.american.edu/news/panama-canal-droughts>, August 30th 2023

³ He is an Associate Dean of Graduate Studies and professor at American University's Kogod School of Business.

⁴ Leslie Jacques, "How Climate Change Is Disrupting the Global Supply Chain", <https://e360.yale.edu/features/how-climate-change-is-disrupting-the-global-supply-chain>, March 10th 2022

⁵ He is a maritime infrastructure resilience scholar at the University of Rhode Island.

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weather, rising sea levels, dwindling agricultural yields, and water scarcity. These perils threatened to disrupt supply chains and production networks on a global scale.⁶

Meanwhile, the Panama Canal, a linchpin in international trade, encountered environmental crisis. Severe drought gripped the region, wreaking havoc on the canal's operations, causing substantial delays and interruptions, as it connects 2,000 ports and 170 countries.⁷ Ivan D. Medina⁸ (Medina) cautioned, "The Panama Canal faces a formidable challenge where numerous vessels encounter obstacles in crossing the passage due to insufficient water levels required for safe transit. The ongoing environmental predicament, marked by a drought during the period when rainfall is typical, has precipitated disruptions in ship navigation through the canal."⁹ Amidst this scenario, the question emerged, could climate change potentially disrupt the Panama Canal, halting the global supply chain?

The Panama Canal: A Gateway to Global Trade

"Global warming and changing weather patterns are affecting the water supply for one of the world's most important waterways, the Panama Canal, as well as access to drinking water for millions of Panamanians", reported Journalist Grace Livingstone from Panama City.¹⁰

In 1914, the Panama Canal, a feat of engineering, connected the Atlantic and Pacific Oceans, revolutionising global shipping. This 82-kilometer artificial waterway replaced the perilous route around Cape Horn¹¹, saving vessels 13,000 kms in transit.¹² Serving as a vital trade artery, the canal managed traffic and tolls for 180 maritime route, benefiting 170 countries.¹³ By 2022, it facilitated transiting over 14,000 ships annually, transporting fuel, grain, minerals, and manufactured goods from East Asia to New York and beyond. Over 40% of goods between northeast Asia and the US East Coast traversed this route.¹⁴ By 2023, the canal handled 2.5% of the total global seaborne cargo.¹⁵ **(Exhibit I)**. Between 35 to 40 ships crossed the canal daily, the numbers fluctuating with water levels. It served as a vital trade route and a major economic asset for Panama.¹⁶ In 2022, \$2.5 billion was contributed to the country's GDP in

⁶ Woetzel Jonathan, et al., "Could climate become weak link in your supply chain?", <https://www.mckinsey.com/~media/mckinsey/business%20functions/sustainability/our%20insights/could%20climate%20become%20the%20weak%20link%20in%20your%20supply%20chain/could-climate-become-the-weak-link-in-your-supply-chain-v3.pdf>, August 2020

⁷ "Unprecedented challenges: Drought at the Panama Canal disrupts global trade", op.cit.

⁸ He is the Professor at Uniminuto of bio-business and internationalization of SMEs.

⁹ Siddiqui Huma, "Navigating Challenges and Innovations: Ensuring the Sustainability of the Panama Canal in a Changing World", <https://www.financialexpress.com/business/defence-navigating-challenges-and-innovations-ensuring-the-sustainability-of-the-panama-canal-in-a-changing-world-3230289/>, September 1st 2023

¹⁰ "Panama Canal grapples with climate change threat", <https://www.bbc.com/news/world-latin-america-62407514>, August 9th 2022

¹¹ "Panama Canal – Mapping", <https://www.insightsonindia.com/2023/08/25/panama-canal/>, August 25th 2023

¹² Yerushalmy Jonathan, "Changing climate casts a shadow over the future of the Panama Canal – and global trade", <https://www.theguardian.com/environment/2023/dec/22/changing-climate-casts-a-shadow-over-the-future-of-the-panama-canal-and-global-trade>, December 22nd 2023

¹³ Swallow Tom, "Panama Canal's Hire Serves Maritime Sustainability Efforts", <https://sustainabilitymag.com/supply-chain-sustainability/panama-canals-hire-serves-maritime-sustainability-efforts>, January 26th 2024

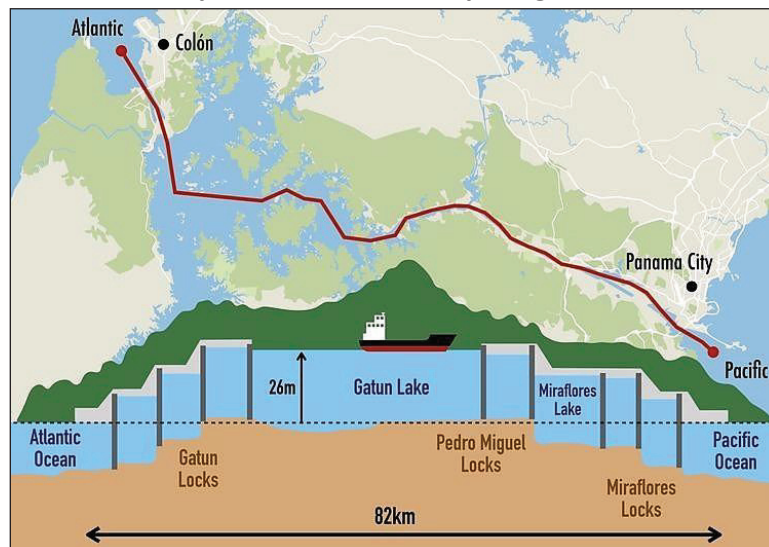
¹⁴ "Changing climate casts a shadow over the future of the Panama Canal – and global trade", op.cit.

¹⁵ Dierker David, et al., "How could Panama Canal restrictions affect supply chains?", <https://www.mckinsey.com/industries/travel-logistics-and-infrastructure/our-insights/how-could-panama-canal-restrictions-affect-supply-chains>, January 19th 2024

¹⁶ Anderson Kara, "How climate change is impacting the Panama Canal", <https://greenly.earth/en-gb/blog/ecology-news/how-climate-change-is-impacting-the-panama-canal>, December 21st 2023

terms of transit fees.¹⁷ Over its history until 2021, the canal contributed around 4 billion Panamanian balboas (equivalent to \$4 billion).¹⁸

Exhibit I
A Map of Panama Canal Depicting its Locks



Source: “The Collapse of the Panama Canal: Disrupting Supply Chains for the Next 10 Years”, <https://www.linkedin.com/pulse/collapse-panama-canal-disrupting-supply-chains-next-10-years-kggyff/>, February 22nd 2024

The Panama Canal functioned through a locking system, employing two sets of locks to elevate and lower ships across its 50-mile stretch. They were raised up to 85 feet (26m) above sea level before being lowered again.¹⁹ Mahelis de García, a Panama Canal Guide explained, “The canal's locks act as a kind of elevator, using enormous amounts of water released from artificial lakes at the top of mountains to raise the vessels.”²⁰ Water from Gatun Lake facilitated this process.²¹ Each ship consumed approximately 55 million gallons of water during transit, which then flowed back into the sea. On average, 37 ships passed through the locks daily using around 2 billion gallons of fresh water.²²

The Canal Climate Cry

The threat of climate change loomed over the Panama Canal, crucial trade chokepoints of the world.²³ Each lock opening discharged millions of gallons of freshwater into the sea, contributing to declining water levels. In 2023, reduced rainfall in Central America was attributed to climate change-induced drought in the canal region. Insufficient replenishment led to unprecedentedly low water levels, hindering the passage of large vessels.^{24,25}

¹⁷ “How could Panama Canal restrictions affect supply chains?”, op.cit.

¹⁸ “Panama Canal’s Hire Serves Maritime Sustainability Efforts”, op.cit.

¹⁹ Donovan John, “How the Panama Canal Makes Water Flow Uphill”, <https://science.howstuffworks.com/engineering/structural/panama-canal.htm>, February 27th 2024

²⁰ “Panama Canal grapples with climate change threat”, op.cit.

²¹ “Changing climate casts a shadow over the future of the Panama Canal – and global trade”, op.cit.

²² “Panama Canal grapples with climate change threat”, op.cit.

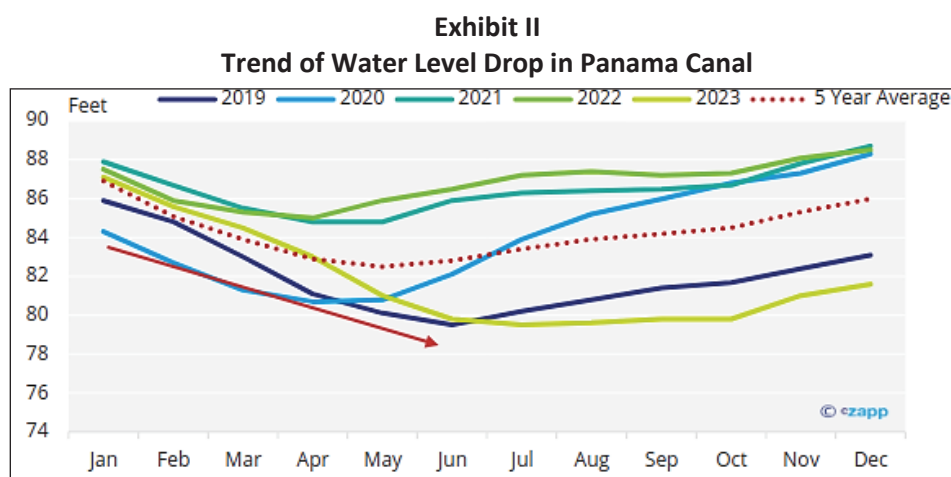
²³ Dahl Hoejris Mie, “The Panama Canal Is Running Dry”, <https://foreignpolicy.com/2024/01/15/panama-suez-canal-global-shipping-crisis-climate-change-drought/>, January 15th 2024

²⁴ Kaufmann Dirk, “Climate change could cut off the Panama Canal”, <https://www.dw.com/en/will-climate-change-cut-off-the-panama-canal-and-global-supply-chains/a-65761965>, May 29th 2023

²⁵ “The Panama Canal Is Running Dry”, op.cit.

Consequently, the Panama Canal Authority (PCA) restricted the number of ships allowed to transit due to near-record low lake levels. Previously, water scarcity was a rare in Panama and was known as one of the wettest countries. However, the El Niño weather pattern caused a rainfall deficit worsening the water levels in Lake Gatun. This resulted in daily water deficit of 3 billion litres, impacting both the canal and local residents.²⁶

According to the PCA (in 2023), “This month of October is the driest since the earliest registers, 73 years ago. The drought caused by the El Niño phenomenon continues to impact the Panama Canal's reservoir system and, as a result, water availability has been reduced.”²⁷ Lake Gatun, a primary water source, reached its lowest level in a decade,²⁸ and was expected to touch 80 feet by April 2024.²⁹ (Exhibit II). Nadim Farajalla³⁰ noted, “Although Panama is used to El Niño, this drought is much more severe than usual. And extreme weather events will likely become more frequent as climate change worsens. We have not yet experienced the full blast of El Niño.”³¹



Source: Warden Sara, “Panama Canal Situation to Get Worse Before It Gets Better”, <https://www.czapp.com/analyst-insights/panama-canal-situation-to-get-worse-before-it-gets-better/>, February 8th 2024

Medina stated, “The reasons for the diminished rainfall vary—some attribute it to human activities, while others perceive it as a manifestation of Earth’s historical fluctuations. However, the consequences are indisputable. This extended dry season has caused water levels in both Gatun and Alajuela lakes to plummet considerably below their seasonal norms, which also jeopardizes the country’s drinking water supply. A substantial demand for fresh water is evident on both sides.”³²

²⁶ “Changing climate casts a shadow over the future of the Panama Canal – and global trade”, op.cit.

²⁷ “Panama Canal to reduce ship traffic due to drought”, <https://news.cgtn.com/news/2023-11-01/Panama-Canal-to-reduce-ship-traffic-due-to-drought-1onyJ7XhaOQ/index.html>, November 1st 2023

²⁸ “Protecting Supply Chains from Panama Canal Drought Disruptions”, <https://www.linkedin.com/pulse/protecting-supply-chains-from-panama-canal-drought-disruptions-37sbf/>, December 6th 2023

²⁹ Warden Sara, “Panama Canal Situation to Get Worse Before It Gets Better”, <https://www.czapp.com/analyst-insights/panama-canal-situation-to-get-worse-before-it-gets-better/>, February 8th 2024

³⁰ She is the Director of the climate change and the environment program at the American University of Beirut’s Issam Fares Institute.

³¹ “The Panama Canal Is Running Dry”, op.cit.

³² “Navigating Challenges and Innovations: Ensuring the Sustainability of the Panama Canal in a Changing World”, op.cit.

Panama's economy relied heavily on natural resources but lacked an adequate environmental framework. The 2015-16 El Niño event prompted a state of emergency, leading to the 'National Water Security Plan 2015-2050' recommended by the previous Government. However, subsequent administrations neglected these efforts.³³ In 2019, a similar incident occurred, marking the fifth driest spell with 20% decrease in rainfall. John Langman³⁴ (Langman), recounted, "In the exceptionally dry year of 2019, the canal authorities had to reduce the amount of water they used to operate the locks, which meant that ships could not carry such heavy cargo because there was less water between the keel of the ship and the bottom of the canal." He emphasised the canal's significance to Panama's economy, stating, "They are working on finding solutions to ensure the canal does not run out of water. We understand that the canal is of huge significance to the Panamanian economy." However, he cautioned, "But it is not only dry years that cause difficulties - heavy rain can also create problems as it can cause the artificial lakes to overflow. As dry years and storms become more common, the canal needs to find fresh sources of water and new ways to store it."³⁵

The water crisis in Panama triggered social unrest in West Panama Province, with residents facing severe water shortages. Protests erupted, mainly targeting the Ministry of Environment due to perceived Government inaction.³⁶ Between 1950 and 2024, Panama's population had quadrupled, with over half relying on the canal's reservoirs for drinking water. Gloria Arrocha Paz³⁷ commented, "Before it was a very small percentage of total water use, and now it's the equivalent of four or five lockages per day."³⁸

The Panama Canal crisis paralleled Europe's 2022 heat waves, causing rivers like the Rhine and Danube to evaporate, disrupting \$80 billion yearly trade. The experts sounded the alarm, expressing deep concerns about the potential paralysis of the global supply chain.³⁹

Will the Panama Canal Foster a Climate-resilient Sustainable Future?

Amid severe drought in August 2023, the PCA cautioned, "The canal and the country face the challenge of the upcoming dry season with a minimum water reserve that must guarantee supply for more than 50% of the population."⁴⁰ The drought led to congestion, with 154 vessels awaiting to cross the passage and extending the waiting time by 21 days.⁴¹ Daily ship

³³ Jordán Osvaldo, "Panama Canal suffers from environmental crisis", <https://latinoamerica21.com/en/panama-canal-suffers-from-environmental-crisis/>, March 22nd 2024

³⁴ He is the Vice President of water projects at the Panama Canal.

³⁵ "Panama Canal grapples with climate change threat", op.cit.

³⁶ "Panama Canal suffers from environmental crisis", op.cit.

³⁷ She is a meteorologist at the Panama Canal Authority.

³⁸ Rojanasakul Mira, "Panama Canal Drought Slows Cargo Traffic", <https://www.nytimes.com/interactive/2024/01/26/climate/panama-canal-drought-shipping.html>, January 26th 2024

³⁹ "Panama Canal traffic is being throttled by climate change impact", <https://economictimes.indiatimes.com/small-biz/sustainability/panama-canal-traffic-is-being-throttled-by-climate-change-impact/articleshow/104957961.cms?from=mdr>, November 4th 2023

⁴⁰ "Drought 5,300 miles away threatens Christmas delivery of iPhones and TVs", <https://kosovotoday.com/drought-5300-miles-away-threatens-christmas-delivery-of-iphones-and-tvs>, December 7th 2023

⁴¹ LaRocco Ann Lori, "'This is going to get worse before it gets better': Panama Canal pileup due to drought reaches 154 vessels", <https://www.cnn.com/2023/08/09/panama-canal-shipping-pileup-due-to-drought-reaches-154-vessels.html>, August 10th 2023

capacity dropped from 36 to 22, later dwindling to 18 by February 2024.⁴² Highlighting the increased likelihood of cargo waiting idle, Captain Adil Ashiq, Head of North America for MarineTraffic commented, “With a reduced transit schedule and an average of 26 daily arrivals by commercial ships per day on the Pacific side of the canal, and an average of 8 daily arrivals by commercial ships per day on the Atlantic side of the canal, the likelihood of cargo waiting idle will increase.”⁴³

As the number of vessels waiting at the canal entrances increased, shipping experts warn of a growing risk of serious accidents. Nitin Chopra (Chopra), Senior Marine Risk Consultant, Allianz Commercial Asia, opined, “Some boats will wait for up to two weeks, anchored at sea on both sides of the canal and they’re running out of space. It’s causing problems with vessels not being able to find a safe spot for anchoring.”⁴⁴

The economic impact of the dipping water levels anticipated by the canal authorities was between \$500 million and \$700 million losses in 2024. Ricaurte Vásquez, Panama Canal Administrator, stated, “In the first quarter of the fiscal year the passageway saw 20% less cargo and 791 fewer ships than the same period the year before.”⁴⁵ The reduction in traffic resulted in an estimated \$200 million drop in the canal’s earnings for 2024.⁴⁶ **(Annexure I).**

The PCA reported, “38% of ships waiting have reservations. The remaining 62% of ships do not, which means they have to wait for the vessels with reservations to proceed across the canal.”⁴⁷ To reduce waiting times, the PCA began booking slots in advance, as it stated, “The Panama Canal urges its customers to make reservations in order to transit as programmed.” However, despite reservations, waiting times extended up to 2.7 days.⁴⁸ These restrictions increased voyage times, leading to higher fuel and freight costs, and raised operational expenses for shipping companies, adding pressure to supply chains and exacerbating post-pandemic recovery challenges.⁴⁹ Oystein Kalleklev, Chief Shipping Executive at Avance Gas revealed, “His company decided in August to pay \$400,000 in a special auction to move a ship ahead in the queue, roughly doubling the total cost of using the canal. Other companies have paid over \$2 million, a cost they will sometimes bear to ensure ships don’t miss their next assignment. A portion of these extra costs will be passed on to consumers, already pummeled by inflation.”⁵⁰ **(Exhibit III).**

⁴² “Changing climate casts a shadow over the future of the Panama Canal – and global trade”, op.cit.

⁴³ LaRocco Ann Lori, “Panama Canal drought hits new crisis level with nearly half of vessel traffic targeted for cuts”, <https://www.cnn.com/2023/11/03/panama-canal-drought-hits-new-crisis-level-amid-severe-el-nino.html>, November 6th 2023

⁴⁴ “Changing climate casts a shadow over the future of the Panama Canal – and global trade”, op.cit.

⁴⁵ “Traffic through the Panama Canal is being slashed because of drought, disrupting global trade”, https://economictimes.indiatimes.com/small-biz/trade/exports/insights/traffic-through-the-panama-canal-is-being-slashed-because-of-drought-disrupting-global-trade/articleshow/106975898.cms?utm_source=contentofinterest&utm_medium=text&utm_campaign=cppst, January 19th 2024

⁴⁶ “Panama Canal to reduce ship traffic due to drought”, op.cit.

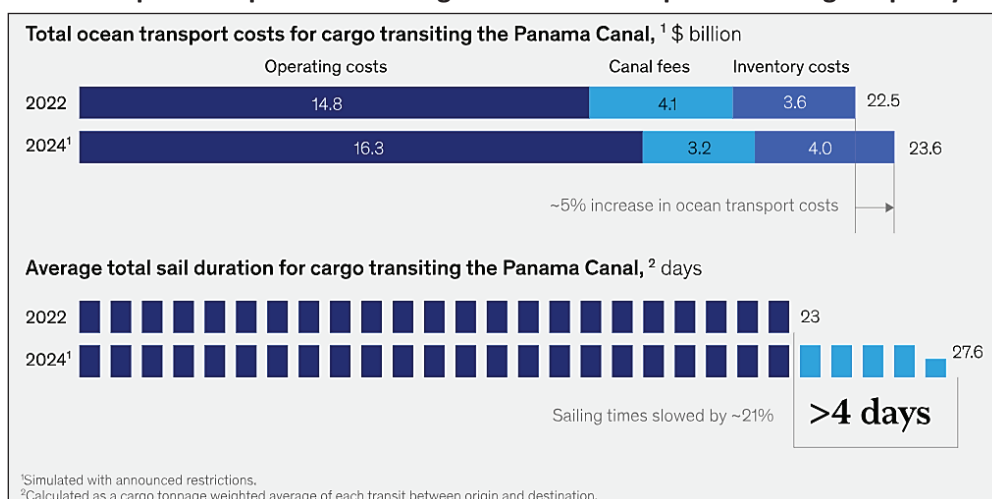
⁴⁷ “‘This is going to get worse before it gets better’: Panama Canal pileup due to drought reaches 154 vessels”, op.cit.

⁴⁸ Murray Christine, “Panama Canal operators to cut ship crossings as drought worsens”, <https://www.ft.com/content/b6604ad4-d2c9-4a00-8a50-1241fa86f26c>, November 1st 2023

⁴⁹ Cabrera Nelson, “Navigating the Panama Canal Crisis: Unraveling the 154-Ship Traffic Jam and Its Global Ramifications”, <https://www.shiplilly.com/blog/navigating-the-panama-canal-crisis-unraveling-the-154-ship-traffic-jam-and-its-global-ramifications/>, August 17th 2023

⁵⁰ Eavis Peter, “Drought Saps the Panama Canal, Disrupting Global Trade”, <https://www.nytimes.com/2023/11/01/business/economy/panama-canal-drought-shipping.html>, November 1st 2023

Exhibit III
A Glimpse of Impact of Reducing Draft Limit of Ships on the Cargo Capacity



Source: Dierker David, et al., “How could Panama Canal restrictions affect supply chains?”, <https://www.mckinsey.com/industries/travel-logistics-and-infrastructure/our-insights/how-could-panama-canal-restrictions-affect-supply-chains>, January 19th 2024

The draft limits⁵¹ of ships passing through the canal were reduced from the normal level of 50 feet to 44 feet, cutting the cargo capacity by 40%.⁵² The PCA warned of further reduction to 43 feet, if water levels hit 78.2 feet.⁵³ This prompted carriers to reduce weight limit and apply a special surcharge between \$300 and \$500 per box. Hapag-Lloyd⁵⁴ issued a customer advisory citing, “The lower-than-usual water levels in the Gatun Lake are causing severe draft restrictions on vessels transiting the Panama Canal. Shippers from East Asia to North America will have to pay a surcharge on all cargo as of June 1.”⁵⁵ The Mediterranean Shipping Company (MSC) had imposed a surcharge of \$297 per container, while Hapag Lloyd levied additional charge of \$130.⁵⁶ The MSC announced, “As a result, pricing for cargo from South Africa, Mozambique & Namibia to South America West Coast transiting the Panama Canal will no longer be inclusive of Panama Canal Surcharge (PCS). Effective 1 January 2024, pricing will be subject to a PCS of USD 200 / TEU.”⁵⁷ (Exhibit IV).

Reduced weight limits for shipping companies resulted in increased costs and scheduling complexities due to recurrent trips.⁵⁸ Nathan Strang (Strang), Director of Ocean Freight at Flexport Inc., a freight forwarder, warned, “The 44-foot draft restrictions are set to reduce

⁵¹ A ship’s draft limit is the height from the seabed to the waterline.

⁵² Laura Curtis, et al., “Panama Canal imposes shipping restrictions to cope with worsening drought”, <https://economictimes.indiatimes.com/small-biz/trade/exports/insights/panama-canal-imposes-shipping-restrictions-to-cope-with-worsening-drought/articleshow/100408508.cms>, May 22nd 2023

⁵³ Laura Curtis, et al., “Why Panama Canal is the new challenge for supply chains”, <https://economictimes.indiatimes.com/small-biz/trade/exports/insights/why-panama-canal-is-the-new-challenge-for-supply-chains/articleshow/100719023.cms?from=mdr>, Jun 3rd 2023

⁵⁴ It is a global leader in container shipping, specialising in reefer cargo, dangerous goods, and special cargo projects.

⁵⁵ “Panama Canal imposes shipping restrictions to cope with worsening drought”, op.cit.

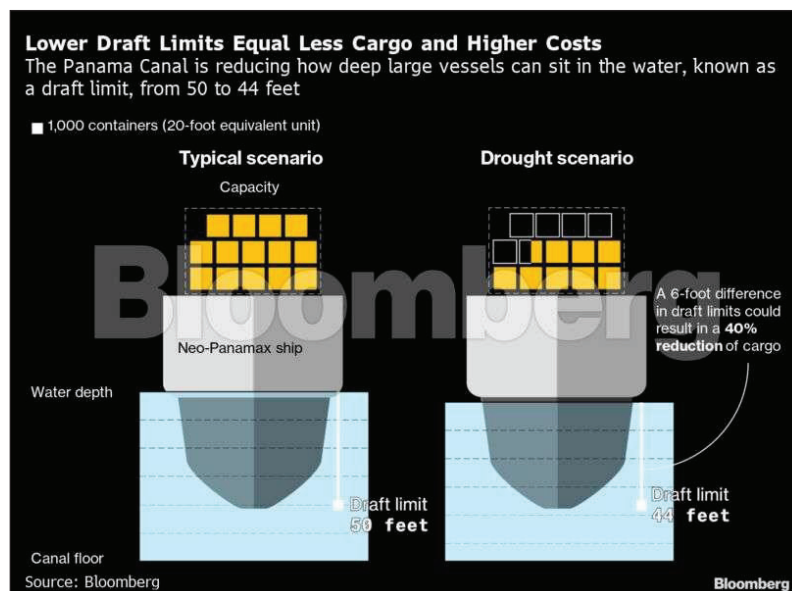
⁵⁶ “Panama Canal Drought: The Impact on Global Shipping Worsens”, <https://www.gep.com/blog/mind/panama-canal-drought-impact-on-global-shipping>, March 4th 2024

⁵⁷ Jeffrey Rebecca, “Panama and Suez Canal double crisis threatens global supply chains”, <https://www.aircargonews.net/business/supply-chains/panama-and-suez-canal-double-crisis-threaten-global-supply-chains/>, December 19th 2023

⁵⁸ Shkuro Sophia, “The Impact of Panama Canal Situation on Global Supply Chains”, <https://www.searates.com/blog/post/the-impact-of-panama-canal-situation-on-global-supply-chains>, January 26th 2024

space for most shippers on Neo-Panamax by 40%. Some shippers will have to split heavier cargo into two containers instead of one, with vessels carrying heavier commodities [getting] affected more. These measures could cost importers and retailers using the route an extra \$1,500 per container.” This led to higher traffic and wait times at the canal, driving up shipment costs. Strang added, “If you ship two containers a week at 12 tons each at roughly \$3,000 apiece then you’ll need to ship three containers. So, your price per shipment went from \$6,000 to \$9,000. And for shippers who typically move 25 tons per container, they’d now need three or four boxes. That’s a much bigger hit and those shippers simply won’t use the canal.”⁵⁹

Exhibit IV
A Glimpse of Impact of Reducing Draft Limit of Ships on the Cargo Capacity



Source: Laura Curtis, et al., “Why Panama Canal is the new challenge for supply chains”, <https://economictimes.indiatimes.com/small-biz/trade/exports/insights/why-panama-canal-is-the-new-challenge-for-supply-chains/articleshow/100719023.cms?from=mdr>, Jun 3rd 2023

Canal fees and long waits heavily impacted shipping companies, straining their competitiveness. Slots were auctioned for nearly \$4 million in November 2023, reflecting the cost pressure.⁶⁰ Ships rerouted their journeys, some from Asia via Cape of Good Hope to the Caribbean, others from South America’s west coast through the Strait of Magellan. Some from North America’s west coast chose the Suez Canal route to the Mediterranean. The rerouting reduced around 3,964 transits through the canal and causing delays and increased operational costs.⁶¹ (Exhibit V).

Suggesting that the impacts of the Suez and Panama Canal crises would be gradual rather than immediate, Pierre Van Der Stichele, VP of Global Cargo at Air Partner, a charter broker, opined, “The affects on the market comes when the peak season is ready with wind down and will not affect the markets from one day to another, but I expect the cost of Ocean freight to increase significantly, whether it be for consumer goods, raw materials, semi-finished goods, built vehicles and outsized cargo, to the point that added sailing time in both instances (ships due to use the Suez and Panama canals), adding several weeks circumnavigating the continents has

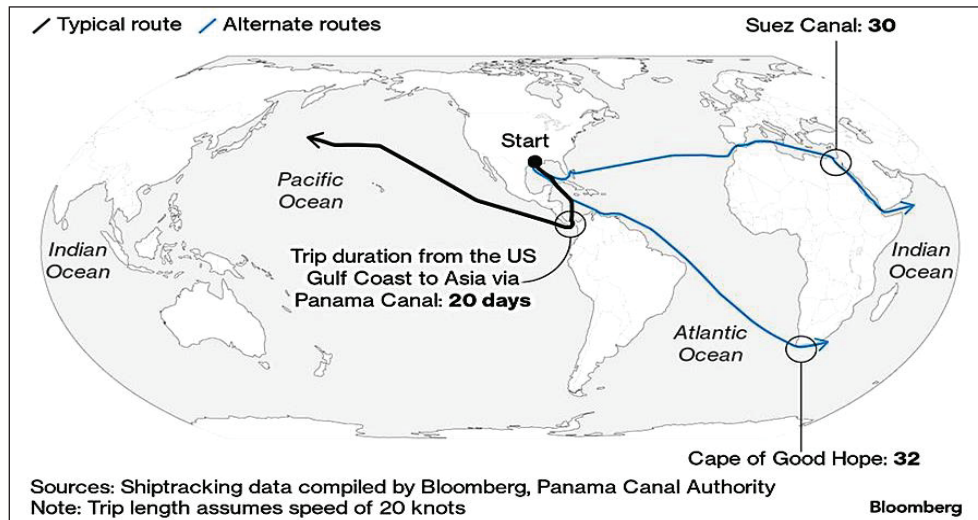
⁵⁹ “Panama Canal imposes shipping restrictions to cope with worsening drought”, op.cit.

⁶⁰ “The Impact of Panama Canal Situation on Global Supply Chains”, op.cit.

⁶¹ “How could Panama Canal restrictions affect supply chains?”, op.cit.

the potential of increasing demand for airfreight just as the low season is due to start which would be welcomed by the air cargo industry.”⁶²

Exhibit V
A Snapshot of Alternative Route Followed by Ships Due to Panama Canal Crisis



Source: Liao Ruth, “Panama Canal jam sends ships sailing continents out of way”, <https://economictimes.indiatimes.com/small-biz/trade/exports/insights/panama-canal-jam-sends-ships-sailing-continents-out-of-way/articleshow/105742809.cms?from=mdr>, December 5th 2023

Maersk and other companies were shifting to the Panama Canal Railway to safeguard supply chains, creating new opportunities for intermodal rail-road routes in Panama.⁶³ José Cervantes⁶⁴ commented, “That unloading and reloading of cargo from ships onto trains and trucks is pushing up costs. And those costs are usually passed onto the consumer.”⁶⁵ “This may start a shift in bookings for Transpacific freight destined to the U.S. East and Gulf through the Suez Canal, which is located in a region with significant geopolitical headwinds,” said Paul Brashier^{66, 67} Chris Rogers, Head of Supply Chain Research at S&P Global Market Intelligence said, “In theory, something that offers a cheaper, shorter route should always be in favour, but it’s the uncertainty that can be a killer.”⁶⁸

S&P Global reported, “Capacity pressures at the Panama Canal are starting to have an effect on supply chains. Container ships have yet to feel the impact in light of their priority status, although the situation is changing.” A major effect has been seen on the flow of bulk carriers.⁶⁹ A huge shift was noticed in the pattern of oil supplies (diesel) from the US, which largely shifted to Europe as South America bought less due to Panama Canal jams and increasing

⁶² “Panama and Suez Canal double crisis threatens global supply chains”, op.cit.

⁶³ Avila Larry, “In Panama Canal droughts, CPKC sees intermodal opportunity”, <https://www.supplychaindive.com/news/cpkc-ceo-sees-intermodal-growth-opportunity-in-panama/705113/>, January 24th 2024

⁶⁴ He is a General Manager at the Panama branch of shipping company Agunsa.

⁶⁵ Fleury Michelle, “Can the Panama Canal save itself?”, <https://www.bbc.com/news/business-68467529>, March 6th 2024

⁶⁶ He is a Vice President of Drayage and Intermodal at ITS Logistics.

⁶⁷ “Panama Canal drought hits new crisis level with nearly half of vessel traffic targeted for cuts”, op.cit.

⁶⁸ “Drought Saps the Panama Canal, Disrupting Global Trade”, op.cit.

⁶⁹ Giraldo Marion, “Panama Canal drought could threaten supply chain, S&P says”, <https://www.reuters.com/business/environment/panama-canal-drought-could-threaten-supply-chain-sp-says-2024-04-03/>, April 4th 2024

freight costs. According to the kpler's data, US transport of diesel to Europe rose from 21% to 45% in December 2023.⁷⁰

A bullwhip effect⁷¹ was experienced by distributors, manufacturers, and retailers, thus creating inventory pressures, longer delivery times, and reorientation of pricing strategies.^{72,73} Jon Davis⁷⁴ commented, "Coal is transported through the canal but the more important energy item is LNG (liquefied natural gas) which the U.S. exports around the world, especially to Asia. Many agricultural products are shipped both from, and to, the U.S. The canal is a major corridor for container ships, so products coming to the U.S., from China for example, are being delayed."⁷⁵ Chopra cautioned, "In the long run we're looking at a big increase in the cost of commodities – it will be passed on to the consumer."⁷⁶

Jan Hoffmann, UNCTAD's Trade and Logistics Head, highlighted, "Here we see the global impact of the crisis, as ships are seeking alternative routes." Rerouted vessels sped up, consumed more fuel, and emitted extra CO₂, worsening environmental concerns.⁷⁷ The crisis also affected the country's economy and had lasting effects on agriculture, biodiversity, and residents of West Panama due to a severe water shortage.⁷⁸

Suggesting the unpredictability in rainfall pattern, Steven Paton⁷⁹ warned, "I certainly am beginning to believe that the patterns we've come to know in the last 30 years are no longer a useful guide in helping us predict the future. Historically there has been a [rainfall] shortage on average once every 20 years due to major El Niño events. In the last 26 years this is the third major rainfall deficit. So it seems that something is changing our rainfall patterns. The increase in frequency is consistent with climate change models, but the climate change models have not come out saying for sure. However, statistically what is going on now has no analogue in the previous 100 years of data."⁸⁰

Highlighting on the escalating frequency, Ilya Espino de Marotta⁸¹ stated, "We depend on rainwater. You used to see a dry year every 15 to 20 years. Now we saw a dry year in 2016, one in 2019, one in 2023, so obviously there's a climate issue we need to address." Erick Cordoba⁸² stated, "He needs to prepare for another drought in the next four years. During the

⁷⁰ Somasekhar Arathy, "Fuel tankers face long slog as Panama Canal drought reroutes flows", <https://www.reuters.com/markets/commodities/fuel-tankers-face-long-slog-panama-canal-drought-reroutes-flows-2023-12-13/>, December 13th 2023

⁷¹ It often occurs when retailers react to demand and amplify expectations around it, causing a domino effect in the supply chain.

⁷² Galipeau Mathieu, "Automation and Predictive Analytics: Panama Canal Case Study", <https://lidd.com/supply-chain-predictive-analytics-panama-canal-case-study/>, December 7th 2023

⁷³ "Shipwrecked: Why everything may soon cost more", <https://zerodha.com/z-connect/subtext/shipwrecked-why-everything-may-soon-cost-more>, March 26th 2024

⁷⁴ He is Chief Meteorologist at global supply chain mapping and risk analytics company Everstream Analytics.

⁷⁵ "Panama Canal drought hits new crisis level with nearly half of vessel traffic targeted for cuts", op.cit.

⁷⁶ "Changing climate casts a shadow over the future of the Panama Canal – and global trade", op.cit.

⁷⁷ "Red Sea, Black Sea and Panama Canal: UNCTAD raises alarm on global trade disruptions", <https://unctad.org/news/red-sea-black-sea-and-panama-canal-unctad-raises-alarm-global-trade-disruptions>, January 26th 2024

⁷⁸ "Panama Canal suffers from environmental crisis", op.cit.

⁷⁹ He is a Scientist at the Smithsonian Tropical Research Institute in Panama.

⁸⁰ "Changing climate casts a shadow over the future of the Panama Canal – and global trade", op.cit.

⁸¹ She is the Panama canal's first Chief Sustainability Officer.

⁸² He is a Manager of the canal's water division and Head of a team of engineers and meteorologists monitoring the canal's watershed.

next drought, drinking water needs will surpass the water available for the transit of ships. That's the problem.”⁸³

Emphasising the need for short-term solutions to mitigate the crisis' impact, Medina stated, “Considering these circumstances, the challenge needs a short-term solution to mitigate the impact of this phenomenon. This involves both waiting for potential shifts in rainfall patterns and embracing innovative approaches, including a form of controlled transformation that addresses the observed scenario. This entails developing new infrastructure and sources for water supply.”⁸⁴

Contrarily, optimists foresee this as a temporary phase. Jorge Luis Quijano, a Consultant and former CEO of PCA quoted said regarding, “The canal could possibly increase to 30 or 32 vessels a day after the dry season ends, and then progressively raise the limit further if rainfall is favorable.” Julia Junnan Zhao, Principal Data Scientist at Dun and Bradstreet, a global data and analytics provider, opined, “Volumes could return to normal in three to five months.” But still the situation created an alarming call for biodiversity. Fred Ogden, a former Professor at the University of Wyoming, emphasised, “Given a big wake-up call to a lot of people. The future does not look bright for the consistency of water resources that we've been able to rely on up until now.”⁸⁵ Providing a perspective on the crisis, Zera Zheng⁸⁶ expressed, “If you put Panama's low water levels into the big picture, actually you see that it's just one signal and globally you will probably see more and more coming.”⁸⁷ The pressing inquiry arose, what measures could the Panamanian authorities undertake to effectively mitigate the water scarcity issue in the Panama Canal?

⁸³ Moloney Anastasia, “‘We all need water': Panama's canal, and people, thirst for more”, <https://www.japantimes.co.jp/environment/2024/03/22/sustainability/water-panama-canal-people/>, March 22nd 2024

⁸⁴ “Navigating Challenges and Innovations: Ensuring the Sustainability of the Panama Canal in a Changing World”, op.cit.

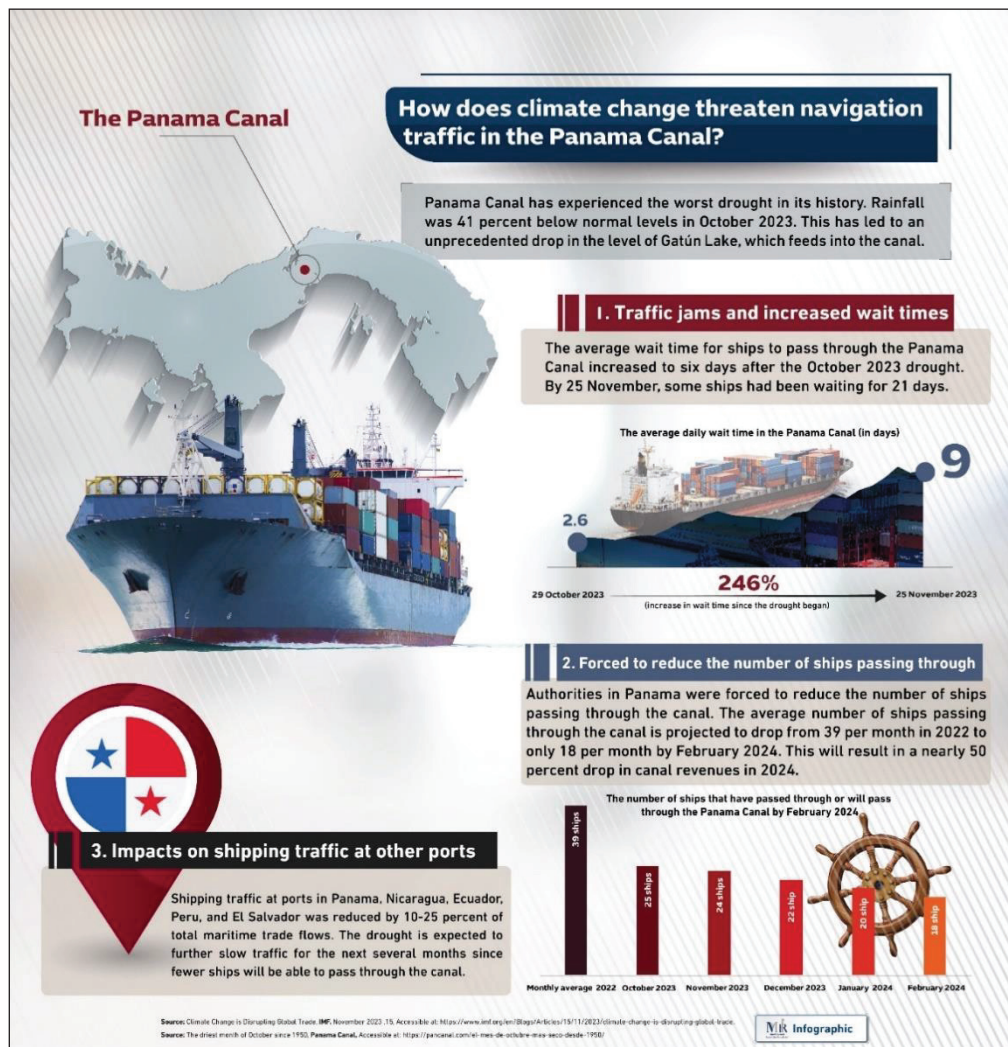
⁸⁵ Millard Peter, et al., “The Panama Canal averts a crisis for now. But at a cost to drinking water”, <https://economictimes.indiatimes.com/small-biz/trade/exports/insights/the-panama-canal-averts-a-crisis-for-now-but-at-a-cost-to-drinking-water/articleshow/108575778.cms>, March 18th 2024

⁸⁶ He is a Head of business resilience consulting at shipping company A.P. Moller-Maersk.

⁸⁷ “Why Panama Canal is the new challenge for supply chains”, op.cit.

Annexure I

An Overview of Trade Disruptions Due to Panama Canal Crisis



Source: "How Does Climate Change Affect Navigation Traffic in the Panama Canal?",
<https://www.interregional.com/article/El-Ni%C3%B1o-Repercussions:/2078/En>, December 5th 2023